SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Computer Vision with OpenCV   COURSE CODE: DSA02

COURSE OUTCOME COVERED
CO2: *Apply image processing and feature extraction techniques, including edge detection, motion estimation, and optical flow, for analyzing images.. (BL3)*

Assessment Tool Weightage: 20%

QUESTIONS: 2   **Total Marks:20**

Annexure A

Class Test

Code of Conduct:

I SK ALI BABU (192524038) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: *SK ALI BABU*

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions	
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation	
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis	
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers	
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps	
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand	

Class Test – Computer Vision Applications :

Question 1: Image Preprocessing and Feature Extraction

INTRODUCTION :

A drone captures farmland images to detect pest-affected regions. Challenges include **noise, shadows, uneven illumination, and rotation/scale variations**. Pest damage appears as **texture and color changes** compared to healthy crops.

a) Preprocessing Pipeline:

Detailed Steps:

1. **Noise Reduction**
   - **Method:** Gaussian or Median Filtering.
   - **Purpose:** Removes random sensor noise caused by wind vibration or sensor limitations.
   - **Impact:** Produces smoother crop textures, making pest damage more visible.
2. **Illumination Correction**
   - **Method:** Homomorphic Filtering / Retinex Algorithm.
   - **Purpose:** Corrects uneven sunlight and shadow effects.
   - **Impact:** Ensures crop reflectance is analyzed independent of lighting conditions.
3. **Contrast Enhancement**
   - **Method:** CLAHE (Contrast Limited Adaptive Histogram Equalization).
   - **Purpose:** Enhances visibility of pest-affected regions by improving local contrast.
   - **Impact:** Highlights subtle differences in crop texture and color.
4. **Normalization**
   - **Method:** Min-Max or Z-score Normalization.
   - **Purpose:** Standardizes pixel intensity ranges across images.
   - **Impact:** Provides consistent input for feature extraction algorithms.

Preprocessing Pipeline

Step	Method	Purpose
Noise Reduction	Gaussian / Median Filtering	Removes random sensor noise and wind distortions for smoother textures.
Illumination Correction	Homomorphic Filtering / Retinex	Corrects uneven sunlight and shadows, highlights true crop reflectance.
Contrast Enhancement	CLAHE (Adaptive Histogram Equalization)	Improves visibility of pest-affected regions without amplifying noise.
Normalization	Min-Max / Z-score	Standardizes pixel intensity range for consistent analysis.

b) Feature Extraction

- **Method:** Gray Level Co-occurrence Matrix (GLCM) or Local Binary Patterns (LBP).
- **Why Suitable:**
  - Pest damage alters **texture** (roughness, irregularity).
  - GLCM captures **spatial relationships of pixel intensities** → detects coarseness and irregularity.
  - LBP encodes **local texture variations** → robust to illumination changes.
- **Outcome:** Healthy crops show uniform texture, pest-affected crops show irregular texture patterns.

c) Rotation and Scale Variations

- **Problem:** Drone captures images at varying heights/angles → features appear rotated or scaled.
- **Effect:** Texture descriptors may fail if not invariant.
- **Solution:**
  - Use **SIFT (Scale-Invariant Feature Transform)** or **SURF (Speeded Up Robust Features)** → robust to rotation & scale.
  - Normalize image size and orientation before analysis.
  - Combine texture features with **color histograms** for added robustness

Question 2: Stereo Vision and Motion Estimation

Introduction:

A self-driving car uses stereo cameras to estimate distances and track moving objects. Challenges include **depth estimation, sudden pedestrian motion, and errors due to occlusion or poor lighting**.

a) Stereo Disparity and Depth Estimation

- Stereo cameras capture **two slightly different views** (left & right).
- **Disparity = pixel shift between corresponding points.**
- **Nearby objects → large disparity.**
- **Distant objects → small disparity.**
- Depth estimated using triangulation:

Depth = (f × B) / disparity

where f = focal length, B = baseline distance.

Stereo Vision Concepts

Concept	Explanation
Stereo Cameras	Capture two slightly different views (left & right).
Disparity	Pixel shift between corresponding points in left vs. right image.
Nearby Objects	Large disparity → appear shifted more.
Distant Objects	Small disparity → almost aligned.
Depth Formula	Depth=f·Bdisparity, where f=focal length, B=baseline.

b) Optical Flow for Motion Tracking

- Optical flow computes **pixel displacement vectors** between consecutive frames.
- If a pedestrian suddenly changes speed/direction:
    - Flow vectors update trajectory.

    - System predicts new motion path.
- **Impact:** Enables real-time tracking and collision avoidance.

c) Errors and Mitigation Techniques

Sources of Error:

- Camera misalignment → incorrect disparity.
- Poor lighting/shadows → false depth.
- Occlusion → missing object parts.
- Fast motion → motion blur.
- Textureless regions → unreliable matching.

Mitigation Techniques:

- **Calibration** with checkerboard patterns.
- **Adaptive histogram equalization** or IR sensors for lighting.
- **Multi-sensor fusion** (LiDAR + cameras).
- **High frame-rate cameras** for fast motion.